of renewable energy and large-scale carbon offset projects, and is committed to achieving net-zero emissions by 2040. It is a signatory to The Climate Pledge, and has set itself science-based emission reduction targets validated by the SBTi.

- HCL, another signatory to The Climate Pledge, is aiming for net-zero carbon emissions by 2040.
- Wipro too targets net-zero by 2040, with an interim 2030 target for 59% reduction in scope 1 and 2 emissions from a 2017 baseline, and a 55% reduction in scope 3 emissions from a 2020 baseline.
- Tech Mahindra is hoping to achieve net-zero across the value chain by 2035, with interim targets validated by the SBTi.
- Zoho is working towards net-zero by 2050, using their own Internet of Things (IoT) technology and other homegrown solutions along the way.
- Tata Consultancy Services (TCS) has set itself a deadline of 2030, while the Tata Group as a whole aims to make its emissions net-zero by 2045.
- LTI Mindtree, a subsidiary of Larsen & Toubro, is shooting for net-zero by 2040, with interim goals like shifting to 85% renewable energy and achieving water positivity by 2030.
- Cognizant is committed to achieving net-zero emissions globally by 2040. Key intermediate targets include a 50% reduction in gross emissions by 2030. By 2026, they hope to supply 100% of their energy needs from renewable sources.
- Bharat Electronics Limited (BEL) is hoping to achieve net-zero scope 1 and 2 emissions by 2030. They were recently awarded the CareEdge-ESG 1 rating with a score of 73.8, for their commitment to sustainable practices.

As global regulations tighten, companies must move beyond carbon neutrality to embrace comprehensive net-zero strategies. The journey requires more than just purchasing carbon credits. It demands fundamental changes in operations, supply chains, and corporate culture. Leading tech companies are showing the way with ambitious targets, innovative solutions, and substantial investments in both technological and nature-based carbon removal. Taking cues from the paths they have charted, build your own roadmap to net-zero emission, because it is no longer a choice but an imperative business transformation. Start now, do it right, and make net-zero your legacy.

Tech-based and nature-based carbon offset methods

There are engineering-based and nature-based approaches to reduce and remove GHGs from the environment. Entities involved in such activities often sell carbon credits in the voluntary market, which many tech companies purchase to accelerate their net-zero targets.

Engineering-based (technological) methods

- **Direct air capture.** Machines filter CO_2 from the air and store it underground for permanent removal, or for use in other products.
- **Bioenergy with carbon capture and storage.** This is a unique CO_2 removal technology that also provides energy, or vice versa. Plants absorb carbon dioxide from the atmosphere during photosynthesis. The biomass from the plants is burned to produce energy. But the CO_2 resulting from the combustion is captured in a concentrated form before it gets to the atmosphere, and stored underground or utilised. The net result is negative emissions.
- **Enhanced rock weathering.** Crushed silicate rocks like basalt are spread over large areas of land to chemically trap CO_2 as stable minerals.
- **Carbon capture, utilisation, and storage.** Captures CO_2 at emission sources for storage or conversion into products.
- **Ocean-based capture.** Removes CO_2 from seawater, thereby increasing the ocean's ability to absorb CO_2 from the atmosphere.
- **Renewable energy projects.** Wind, solar, and geothermal replace fossil fuels and prevent or reduce CO_2 emissions.
- **Energy efficiency.** Upgrading industrial and domestic systems to reduce emissions.
- **Storing and reusing CO_2.** Conversion of captured CO_2 into usable products also generates carbon credits.

Nature-based methods

- **Afforestation/reforestation.** Planting/restoring forests, especially mangroves, helps absorb atmospheric CO_2
- **Biochar.** Organic waste is converted into stable carbon (charcoal), which is locked into soil.
- **Soil sequestration.** Improved land use and farming (cover crops, no-till) boost organic carbon in soil.
- **Blue carbon initiatives.** Restoring/protecting mangroves, seagrass, and coastal wetlands enable rapid, durable carbon storage.
- **Peatland restoration.** Re-wetting peatlands prevents oxidation and large CO_2 emissions.
- **Avoiding deforestation.** Protecting forests from being cleared prevents GHG emissions. Reducing Emissions from Deforestation and Forest Degradation (REDD+) is a framework under the United Nations Framework Convention on Climate Change to incentivise developing countries to protect forests and mitigate climate change.
- **Improved forest management.** Better management of existing forests also increases carbon stocks and enhances ecosystem services.
- **Rice methane reduction.** Changing rice farming practices reduces methane emissions.
- **Developing micro forests.** Using the Japanese Miyawaki method to create and maintain dense forests in small spaces, even in urban areas, helps capture carbon dioxide and improve air quality.

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